

ALI NIKKHAH

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RESEARCH INTERESTS

Vision-language models; multimodal representation learning; explainable and robust AI; speech and audio intelligence; applied machine learning for healthcare and financial systems.

EDUCATION

Sharif University of Technology

B.Sc. Electrical Engineering – Digital Systems GPA: 3.65 overall, 3.91 major

• Relevant coursework: Deep Learning, Machine Learning, Digital Signal Processing, Linear Algebra, Probability & Statistics, Algorithm Design, Digital Systems Design.

Tehran, Iran

Sept 2020 – Jul 2024

PUBLICATIONS & PREPRINTS

Automated DICOM-Compliant Ultrasound Report Generation via Contrastive Vision-Language Learning

2025

Ali Nikkhah, [Co-authors TBD], **A. Supervisor**

arXiv preprint – in preparation

RESEARCH EXPERIENCE

University of British Columbia

Research Assistant – Vision-Language Medical Imaging

- Developed vision-language architectures (**ViT**, **BLIP**, **T5**) trained with self-supervised contrastive learning to generate DICOM-compliant diagnostic reports from raw ultrasound scans; achieved SOTA **BLEU** and **ROUGE-L** on the evaluation benchmark.
- Investigated cross-modal alignment strategies to bridge the semantic gap between ultrasound image features and structured radiology report language.
- Work resulted in a preprint in preparation (see Publications).

Trinity College Dublin – EmoDub Project

ML Researcher – Emotion-Aware Voice Translation

- Researched emotion-preserving voice translation, developing architectures that maintain both linguistic content and affective characteristics across language boundaries.
- Fused acoustic, semantic, and contextual embeddings using **wav2vec2.0** and **Whisper** to represent speaker emotion independently of lexical content, enabling emotionally faithful dubbing generation.

L3S Research Center (Leibniz University Hannover)

Research Assistant – Video Motion Classification

- Designed texture-free motion classification pipelines using 3D CNNs and transformer encoders applied to dense optical flow fields, deliberately suppressing appearance cues to isolate generalizable human-dynamic representations.
- Achieved state-of-the-art classification accuracy on **UCF-101** and **HMDB-51** benchmarks; findings demonstrated that flow-based texture-agnostic features transfer more robustly across recording conditions.

Sharif University of Technology – Applied ML Lab

Research Collaborator – Explainable AI

- Investigated domain-agnostic Explainable AI (XAI) frameworks, applying advanced attention visualization techniques to probe the internal representations of large neural networks.
- Explored robustness of ML model explanations under distribution shift, contributing to lab discussions on interpretability evaluation methodology.

TEACHING & MENTORING

Sharif University of Technology

Graduate & Undergraduate Teaching Assistant

- Foregrounded courses (graduate-level):** Deep Learning, Natural Language Processing, Data Science, Machine Learning.
- Additional courses (undergraduate-level):** Generative Models, Engineering Probability & Statistics, Analog Electronics, Circuit Theory.
- Responsibilities across roles included designing and grading problem sets, holding weekly office hours for cohorts of 30–80 students, preparing tutorial sessions, and contributing to exam question design for graduate-level deep learning and NLP courses.
- 3 years continuous TA service across 8 distinct courses spanning both core electrical engineering and advanced AI/ML curricula.

INDUSTRY EXPERIENCE

Digikala <i>AI Engineering R&D Lead</i>	Tehran, Iran Nov 2024 – Aug 2025
• Led applied research into agentic RAG architectures, implementing Knowledge-Augmented Generation (KAG), hybrid dense/sparse retrieval (RRF), and HyDE query expansion in a production system serving 5M+ daily queries. • Investigated multi-agent system design using LangGraph: modeled tool-calling agents as cyclic directed graphs with explicit state transitions, evaluating tradeoffs between agent autonomy and output reliability.	
Advanced Analytics Australia <i>Computer Vision Engineer</i>	Remote Contract Sep 2025 – Apr 2026
• Applied deep vision models (ViT, object detection, optical flow) to real-world safety compliance; investigated quantization and runtime compilation (TensorRT, TF Lite) for constrained edge deployment.	
Turquoise Digital (Firouzeh Digital) <i>Senior Data Scientist & Data Engineer</i>	Tehran, Iran Aug 2025 – Mar 2026
• Applied time-series transformer models to financial forecasting; designed agentic chatbot with function-calling for secure live data retrieval; built telephony audio transcription pipeline using Whisper and wav2vec2.0.	
HomaCloud <i>MLOps Engineer</i>	Tehran, Iran May 2021 – Feb 2022
• Orchestrated distributed ML pipelines on Kubernetes (Ray, MLflow, TensorFlow Serving); managed CI/CD and infrastructure-as-code (Terraform, ArgoCD).	

TECHNICAL SKILLS

Languages: Python, C++, Go, SQL, R, Bash, TypeScript
Deep Learning & Vision: PyTorch, TensorFlow, JAX, Hugging Face, ViT, BLIP, T5, Whisper, wav2vec2.0, Librosa
Agentic AI & RAG: LangGraph, LangChain, LlamaIndex, FAISS, ChromaDB, HyDE, KAG, Graph RAG
Data & MLOps: Spark, Airflow, Kafka, Docker, Kubernetes, Ray, Triton, vLLM, MLflow, W&B
Languages (Human): Persian (Native), English (C2 – IELTS 8.0), Arabic (Professional), French (Beginner)